

CANDIDATE NAME: _____

INDEX NUMBER _____

CG _____



SERANGOON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2018

H1 BIOLOGY

Paper 2 Structured and Free-response Questions

8876/02

Thursday
13 September 2018

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, index number and CG in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE	
Paper 1 (MCQ)	/30
Paper 2	
Section A	
1	/6
2	/8
3	/11
4	/9
5	/11
Section B	
6 or 7	/15
P2 Total	/60
Total	/90
TOTAL (100%)	

Section A

Answer **all** the questions in this section.

1. Fig. 1.1 is an electron micrograph of part of an animal cell.

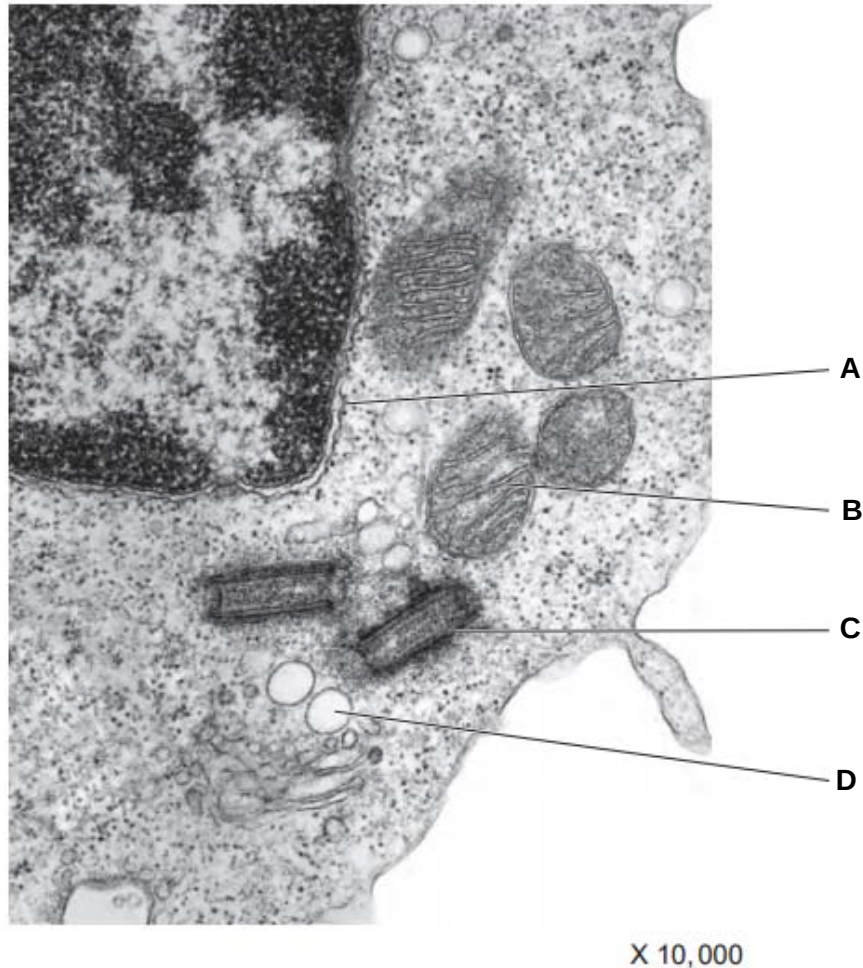


Fig. 1.1

- (a) Name the membranes labelled **A** and **B**. [2]

A: nuclear envelope

B: crista(e)/ inner mitochondrial membrane

- (b) Describe the role of structure **C** in cell division. [2]

1. Assembly/ organisation of spindle fibres/ microtubule
2. To separate chromosomes/ chromatids during mitosis/ meiosis/ cell division

(c) Explain how it possible for the pH in structure **D** to be kept constant. [2]

1. Membrane of structure D is largely hydrophobic due to hydrophobic fatty acid tails of phospholipid bilayer
2. Prevents charged/ hydrophilic H⁺ (*Reject: polar*) from diffusing across freely

[Total: 6]

2. Fig. 2.1 is an electron micrograph of a cancer cell in the midst of mitosis.

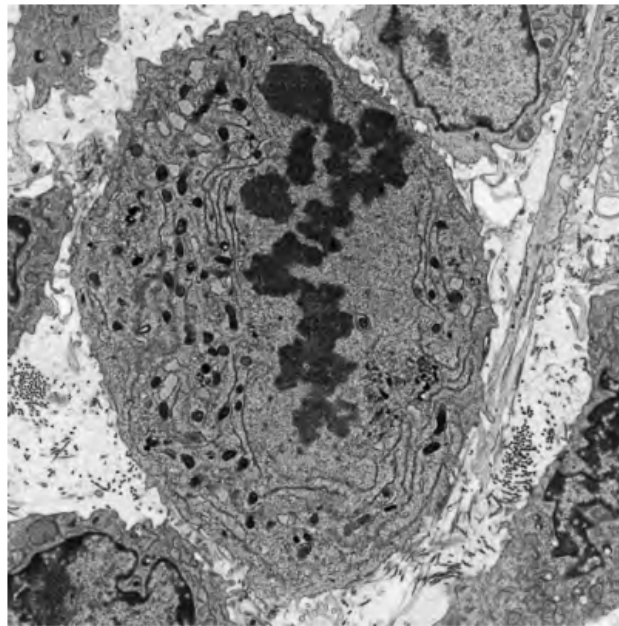


Fig. 2.1

(a) Identify the stage of mitosis, giving a reason for your answer. [2]

1. Stage: Metaphase
2. Reason: chromosomes align in a single/ one row at the equator/ metaphase plate

People who have smoked cigarettes for many years are at risk of developing lung cancer.

(b) Explain why long-term smoking increases the risk of cancer in smokers. [2]

1. Cigarettes contain chemical carcinogens/ or named example e.g. tar, nitrosamines which increases the rate of mutation
2. Ref. to long-term smoking and (idea of) continued exposure to carcinogens, hence leading to the accumulation of mutations that cause cancer

Fig. 2.2 shows the change in the percentage of smokers in the male population of the UK between 1950 and 2005. Fig. 2.3 shows the change in mortality rate in the UK in men aged 75 to 84 between 1950 and 2005.

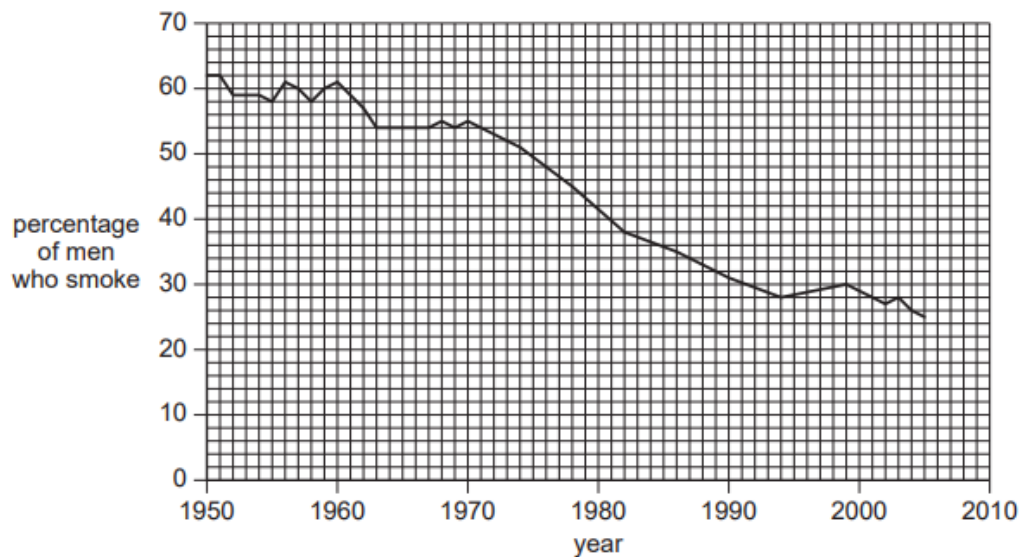


Fig. 2.2

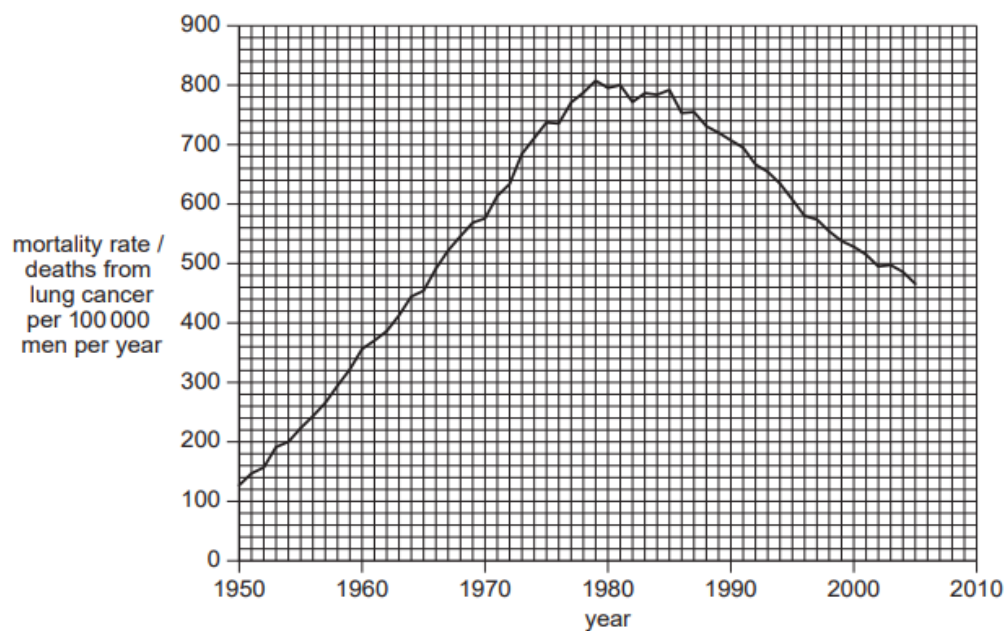


Fig. 2.3

(c) With reference to Figs. 2.2 and 2.3, discuss the observations made between 1950 and 2005. [4]

1. Percentage of men who smoke fell from 62 to 25% between 1950 and 2005
2. and mortality rate increased from 130 to 810 deaths from lung cancer per 100 000 men per year between 1950 and 1979 and then decreased to 470 in 2005
3. Increase in number of deaths despite fall in % smokers between 1950 and 1979 as cancer is multistep process/ requires multiple mutations to accumulate before it

develops into cancer hence lag time/ death from cancer is often associated with metastasis which occurs typically in the later stages of cancer development

4. fall in cancer deaths due to improvements in health care that allow for earlier diagnosis/ treatment/AVP

[Total: 8]

3. DNA and RNA are some of the key molecules involved in the synthesis of proteins.

(a) Comment on how DNA and tRNA are different, in terms of the bases present.

	DNA	tRNA
bases	Thymine present	Thymine absent, uracil is used instead
A:T/U G:C ratio	A:T and G:C ratio is always 1	A:T/U and G:C ratio varies

(b) Calculate the minimum number of DNA nucleotides required to code for a polypeptide with 238 amino acids.

- Show your working in the space below.
- State the **two** properties of the genetic code that are considered in deriving your answer. [3]

1. $238 \times 3 = 714$

2. Stop codon: 3 bases, Total: $714 + 3 = 717$ DNA nucleotides

Property 1: The genetic code is a triplet code.

Property 2: The genetic code is punctuated/ contains stop codons.

(c) Describe the role played by tRNA in polypeptide synthesis. [3]

1. Carries amino acid to tRNA binding sites/ A site on ribosome
2. Brings growing polypeptide chain and incoming amino acid in close proximity for formation of peptide bond
3. Specific amino acid sequence ensured by complementary base pairing of anticodon on tRNA with codon on mRNA

- (d) Regulation can occur following the synthesis of proteins. An example of such regulation is shown in Fig. 3.1, where inactive enzyme precursor pepsinogen is modified post-translationally to form active pepsin.

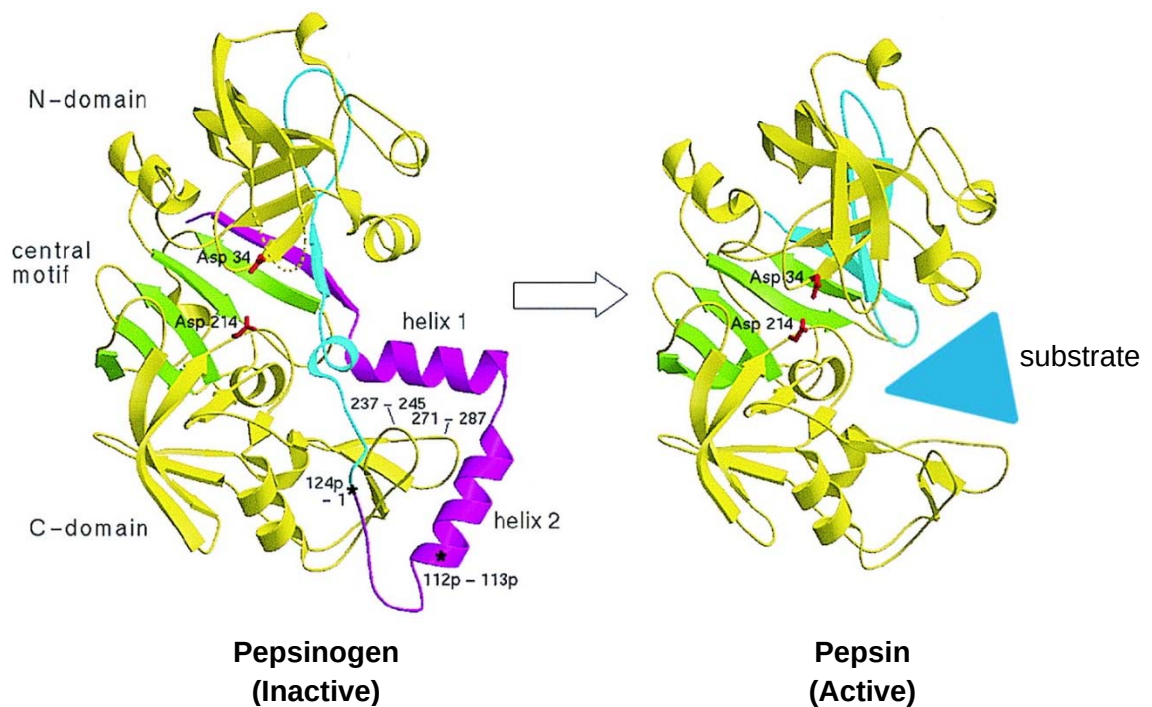


Fig. 3.1

- (i) With reference to Fig. 3.1, describe the post-translational change observed in the enzyme and explain why it results in its activation. [2]
1. Proteolytic cleavage/ activation of pepsinogen which removes helix 1 and 2
 2. Exposes the active site to allow for substrate to bind and pepsin to catalyse breakdown of proteins
- (ii) Suggest why pepsin is synthesised as an inactive precursor pepsinogen. [1]
1. (Idea of) preventing pepsin from prematurely digesting protein components of the cells that synthesise them

[Total: 11]

4. The fruit fly, *Drosophila melanogaster*, has eyes, a striped abdomen and wings longer than its abdomen. This is called a 'wild-type' fly.

Mutation has resulted in many variations of these features. Table 4.1 shows diagrams of a wild-type fly and three other flies, each of which shows one **recessive mutation**.

Table 4.1

				
eyes	present	present	absent	present
abdomen	striped	black	striped	striped
wing description	long	long	long	short

(a) Explain what is meant by a "recessive mutation". [2]

1. Change in DNA/ nucleotide sequence leading to the creation of a new allele, whose
2. expression is masked in the presence of a dominant/ normal allele OR only expressed in the presence of another recessive/ mutated allele

(b) Using appropriate symbols, illustrate a cross between a fly without eyes and long wings and one with eyes and short wings that would result in **four** different phenotypes being observed in the offspring. [4]

Let E represent the dominant allele for presence of eyes and e the recessive allele for absence of eyes

Let L represent the dominant allele for long wing and l the recessive allele for short wings

Parental phenotype: Without eyes, long wings x With eyes, short wings

Parental genotype: [1] eeLl x Eell

Gametes: [1] eL el x El el

Punnett square: [1]

	eL	el
El	EeLl	Eell
el	eeLl	Eell

Offspring genotype: EeLl: Eell: eeLl: eell

Offspring phenotype: [1]
 with eyes, long wings:
 with eyes, short wings:
 without eyes, long wings:
 without eyes short wings

Phenotypic ratio 1:1:1:1

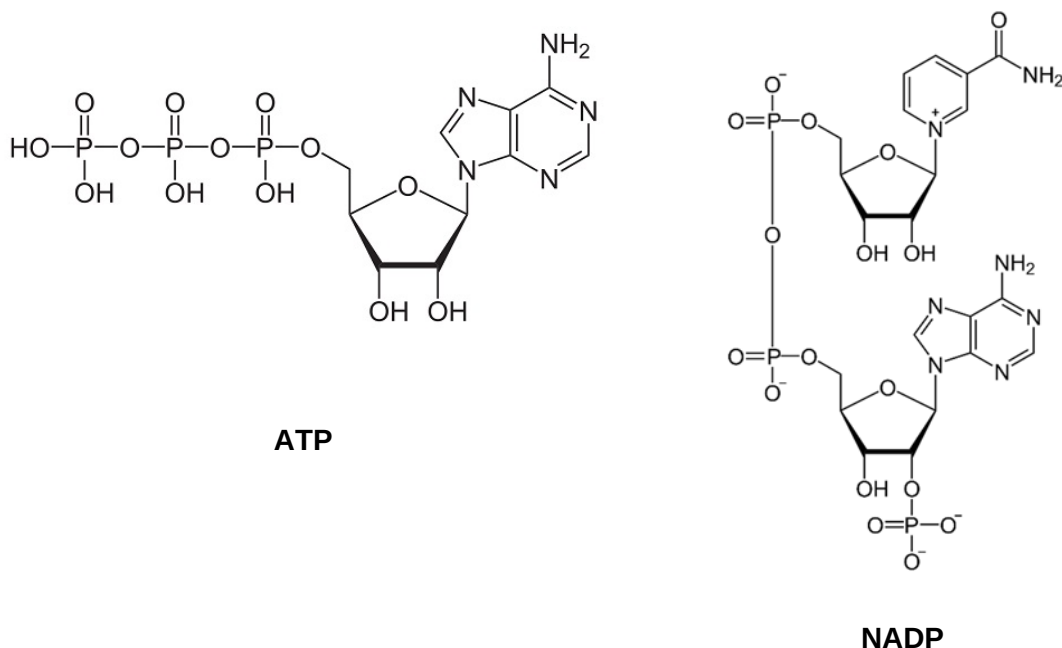
- (c) The class of mutations giving rise to the various mutant phenotypes in Table 4.1 are termed as **gene mutations**.

Explain how these mutations are different from chromosomal aberrations. [3]

Gene mutation	Chromosomal structural mutation
1. Changes in DNA/nucleotide sequence of a gene	- Changes in chromosome structure/ involves exchange of chromosome segments - DNA/nucleotide sequence of gene (mostly) unchanged
2. Alters a single gene locus on a chromosome	Alters more than one loci on chromosomes
3. Caused by deletion, insertion, substitution or inversion of one/ several nucleotides	Deletion, inversion, translocation or duplication of chromosomal fragments/ several gene loci
4. Give rise to new alleles	Rearrangement of loci of genes / alleles / new combination of alleles
AVP:	
5. May change amino acid sequences or form non-functional proteins/ silent, missense, nonsense mutation	Amino acid sequences usually unchanged but changes the level of expression of genes / inactive, hyperactive, underproduction or overproduction of gene product
6. Play more important role in evolution than chromosomal mutations because new alleles increases variation in the gene pool for natural selection to operate	Play a less important role in evolution than gene mutations because chromosomal mutations involve only reshuffling of alleles that already exist in gene pool

[Total: 9]

5. ATP and NADP both play important roles in photosynthesis. Fig. 5.1 represents the



molecular structures of ATP and NADP.

Fig. 5.1

(a) Using Fig. 5.1, compare the structures of ATP and NADP. [3]

1. Both have three phosphate groups
 2. Both have an adenine/ purine
 3. Both have pentose sugar/ ribose
 4. Both have phosphoanhydride bonds
- Max 2

5. ATP has one pentose sugar while NADP has two
 6. NADP has nicotinamide base which is absent in ATP
 7. There are 2 phosphoanhydride bonds present in ATP, but only one present in NADP.
- Max 2

(b) Outline the roles of NADP in a cell. [2]

1. Serves as final electron acceptor in non-cyclic photophosphorylation
2. To form reduced NADP/ NADPH which is used in light-independent reaction/ Calvin cycle

(c) State the names of the processes in which ATP is synthesised during photosynthesis and respiration respectively. [2]

1. Photosynthesis: cyclic and noncyclic photophosphorylation
2. Respiration: substrate level phosphorylation and oxidative phosphorylation

- (d) ATP serves as a source of energy for several metabolic processes in both photosynthesis and respiration.

Name one process in **respiration** that requires ATP as an energy source. [1]

1. Active transport of pyruvate into mitochondrion
2. Phosphorylation of intermediate compounds in glycolysis
3. AVP

- (e) The first substrate used in respiration is glucose. In a situation of excess glucose, some of these glucose is stored as fats instead of carbohydrates.

Explain why animals prefer to store lipid instead of carbohydrates. [3]

1. Twice the amount of energy produced per gram/ unit mass of lipid/ carbohydrate stored due to higher proportion of C-H to O present
2. Twice the amount of metabolic water per gram/ unit mass of lipid/ carbohydrate oxidised due to higher proportion of C-H to O present
3. Smaller volume stored for the same amount of energy hence aiding in locomotion
4. Ref. to lipids (triglycerides) being good thermal/ heat insulators
5. Ref. to lipids being stored around delicate organs hence acting as cushioning material/ protecting these organs
6. Ref. to lipids being less dense than water, thus providing buoyancy for aquatic animals

Max 1 for points 4-6

[Total: 11]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)** and **(b)**, as indicated in the question.

7. **(a)** Explain how climate change affects the spread of dengue beyond the tropics. [5]

- (b)** Discuss how variation is generated in a population, and explain why it is important in natural selection. [10]

[Total: 15]

8. **(a)** Explain how man's food choices can lead to climate change impacts. [5]

- (b)** Explain what is meant by the fluid mosaic model, and discuss the importance of membrane fluidity in biological processes. [10]

[Total: 15]

6 (a) Explain how climate change affects the spread of dengue beyond the tropics. [5]

1. Ref. to climate change causing **global warming/ increasing global temperatures** leading to **sub-tropical areas/ higher latitudes** experiencing **warmer temperatures**
2. Ref. to dengue being spread by the female ***Aedes aegypti*** mosquito
3. Which acts as the **vector** for the virus
4. (Idea of) warmer temperatures in sub-tropical areas being **more optimal/ conducive** for mosquito vector
5. in terms of **higher rate of development/ higher metabolic rate/ higher rate of survival** of mosquitoes
6. Ref. to climate change leading to **increased rainfall**, increased pools of **stagnant water/ breeding sites** for mosquitoes in sub-tropical regions.
7. Accept AVP, e.g. spread of dengue beyond the tropics is compounded by increased international travel/ population growth

6 (b) Discuss how variation is generated in a population, and explain why it is important in natural selection. [10]

1. **Meiosis** results in genetic variation
2. ... due to **crossing over between non-sister chromatids of homologous chromosomes**, during **prophase I**;
3. where **homologous regions/ regions containing same genes** are **exchanged at chiasmata**;
4. creating **new combination of alleles** (**Reject: form new genes**);
5. ... due to **independent assortment of homologous chromosomes** during **metaphase I**;
6. whereby the **orientation of homologous pairs of chromosomes/ bivalents** along the **equator/ metaphase plate** is **random/ independent** of other bivalents;
7. Ref. to daughter cells having **equal probability of receiving either the paternal or maternal copy** of each chromosome
8. **Independent assortment of chromosomes/ pairs of chromatids** also occurs during **metaphase II**, **significant due to crossing over** which occurred during prophase I;
9. In addition, during **fertilisation**, **random** fusion of gametes occurs, resulting in numerous combinations of chromosomes in a zygote.
10. Gene **mutations** contributes **new alleles** to the gene pool
11. Ref. to how gene mutations are caused, i.e. due to **mistakes in DNA replication**, due to **mutagens** (or **named example**)

Max 6

12. Genetic variation is the **raw material** for natural selection to occur
13. Ref. to a wide enough **range of phenotypes/ alleles** present in a population allowing for **selection pressure** to act on
14. Leading to **differential reproductive success**
15. Such that **individuals with advantageous phenotypes will survive to reproductive age and pass the alleles** that confer these advantageous phenotypes on to their **offspring**
16. Leading to **change in allele frequencies** in gene pool **over time**

Max 4

QWC: At least 2 causes of variation discussed, and why variation is important for natural selection systematically explained.

7 (a) Explain how man's food choices can lead to climate change impacts. [5]

1. Ref. to **increased consumption of/ demand for meat/ dairy products**
2. Leading to **more cattle/ cows** being reared as livestock
3. Ref. to cows being ruminant mammals which **release methane**
4. Ref. to methane being a **greenhouse gas** which **traps heat in the earth's atmosphere**
5. Causing **increasing global temperatures/ global warming**
6. **Clearing of forested land/ deforestation** to create **land for grazing** leads to **increased carbon dioxide emission/ reduced carbon sinks**
7. Accept AVP, e.g. increasing affluence leading to increased meat demand

7 (b) Explain what is meant by the fluid mosaic model, and discuss the importance of membrane fluidity in biological processes. [10]

1. Fluid refers to the **constant lateral movement** of **phospholipids** and other membrane components, e.g. membrane proteins
2. Due to **weak hydrophobic interactions** between fatty acid tails of phospholipids
3. Mosaic refers to the **random arrangement** of **membrane proteins** in bilayer
4. Named example of membrane protein, e.g. receptor, transport protein, aquaporin

Max 3

Importance of membrane fluidity in transport processes

5. Ref. to membrane fluidity creating **transient gaps** in phospholipid bilayer
6. Allowing **small/ uncharged/ non-polar/ water molecules** to **freely diffuse** across
7. Ref. to membrane fluidity allowing **fusion of vesicles** with endoplasmic reticulum in the **secretory pathway/ in intracellular transport of materials**
8. E.g. **fusion of transport vesicle with Golgi apparatus** + outcome, i.e. allowing proteins to undergo **chemical modification in the GA**
9. Ref. to **fusion of lysosome with food vacuole** + outcome, i.e. allowing for **digestion of ingested bacteria**
10. Ref. to **exocytosis/ fusion of secretory vesicles with the cell surface membrane** + outcome, i.e. **release of secretory proteins**, localisation of **transport proteins on cell surface membrane**
11. Ref. to importance of membrane fluidity in **endocytosis**
12. Where **cell surface membrane invaginates** to form endocytic **vesicles/ food vacuoles**
13. Hence allowing cell intake of **large quantities** of extracellular substances

Max 7

Importance of membrane fluidity in cell division

14. Allows for formation of **cleavage furrow** in animal cells
15. Thus facilitating **cytokinesis**

Max 2

QWC: Fluid mosaic defined + at least 2 named examples of how instances of membrane fluidity are evident in the cell AND importance of these processes discussed